Physics 12 Winter 2026
Week 6 Discussion

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1. Summary of Topics from Last Week

- *From the lectures*
  - ◇ Atomic structure, atomic number (Z), atomic mass (A), isotopes
    - ∗ Atomic number = number of protons/positive charges in nucleus. Atomic mass = number nucleons (protons and neutrons).
  - ◇ Fundamental forces (electromagnetic, gravitational, strong, weak)
  - ◇ Radioactivity, decay (alpha, beta, gamma), half life, ionizing radiation
    - ∗ Alpha decay emits a alpha particle (helium nucleus/$Z = 2, A = 4$). Beta decay emits a beta particle (electron/$Z = -1, A = 0$). Gamma decay emits a high energy photon ($Z = 0, A = 0$).
  - ◇ Radiation dosage (units), application of radioactivity
  - ◇ Nuclear reactions
    - ∗ Rules: Atomic number Z is conserved. Atomic mass A is conserved.
  - ◇ Mass-energy equivalence ($E = mc^2$), rest energy, units (eV, amu)
  - ◇ Nuclear binding energy
  - ◇ Nuclear fission, fissile material, chain reaction, critical mass
    - ∗ A spherical shape minimizes the critical mass because it minimizes leakage of neutrons by minimizing the surface-area-to-volume ratio.
  - ◇ Uranium enrichment, nuclear reactor, breeder, coolant, moderator
  - ◇ Nuclear fusion (deuterium + tritium)
- *From the textbook* (Wolfson 4th edition)
  - ◇ Atomic nucleus, isotopes (Chapter 7.1)
  - ◇ Radioactive decay (Chapter 7.2)
  - ◇ Nuclear binding energy (Chapter 7.3)
  - ◇ Nuclear fission, chain reaction, critical mass (Chapter 7.4)
  - ◇ Nuclear reactors (LWR, breeder) (Chapter 7.5)
  - ◇ Uranium enrichment, nuclear waste (Chapter 7.6)
  - ◇ Environmental and health impacts (radiation dose) (Chapter 7.7-7.8)
  - ◇ Nuclear fusion (Chapter 7.9)

2. **Brain-warmer.**

   (a) **Radiation source.** Which of the following is the biggest source of radiation for the human body?

   A. Radon gas
   B. Cosmic rays
   C. Potassium-40 and Carbon-14 from food
   D. Medical procedures (including CT scans and X-rays)
   E. Contamination from nuclear accidents

   Answer: A.

   (b) **Criticality.** Which of the following factors determine the critical mass of a radioactive material?

   A. The geometric shape of the material
   B. The concentration of fissile isotopes
   C. Both of the above
   D. None of the above

   Answer: C.

3. **Radioactive decay.**

   (a) **Radon-222 decay chain.** The radioactive radon gas undergoes a decay chain just like any other radioactive elements. One of the possible decay chain for radon-222 is

$$^{222}_{86}\mathrm{Rn} \overset{\text{alpha}}{\rightarrow} \mathrm{Po} \overset{\text{beta}}{\rightarrow} \mathrm{At} \overset{\text{alpha}}{\rightarrow} \mathrm{Bi} \tag{1}$$

   What are the atomic numbers Z and atomic masses A of the decay products Po, At, and Bi?

   Solution: In an alpha decay, the nucleus emits an alpha particle/helium nucleus ($Z = 2, A = 4$), so the atomic number of the nucleus should decrease by 2 and the atomic mass should decrease by 4. In a beta decay, the nucleus emits a beta particle/electron ($Z = -1, A = 0$), so the atomic number of the nucleus should increase by 1 and the atomic mass should stay constant. Therefore, the resulting isotopes are $^{218}_{84}\mathrm{Po}$, $^{218}_{85}\mathrm{At}$, and $^{214}_{83}\mathrm{Bi}$.

(b) **Cesium-137 decay.** ^{137}Cs is a waste product from fission reactors. The half-life of ^{137}Cs is 30.1 years. (i) How long does it take for the activity of a Cs-137 sample to drop by a factor of 3? (ii) By what factor would the activity drop if three times as much time as the answer for the last question has passed?

Solution:

i. We can describe the activity at time t (measured in years) using $A(t) = A_0 \cdot 2^{-t/t_{1/2}}$, where A_0 is the activity at $t = 0$ and $t_{1/2}$ is the half-life. When the activity drops by a factor of 3, we have

$$\frac{A_0}{A(t)} = 2^{t/t_{1/2}} = 3 \tag{2}$$

so $t = t_{1/2} \log_2 3 = 47.7$ years.

ii. After each 47.7 years, the activity would drop by a factor of 3, so after 3 periods of 47.7 years, the activity would drop by a factor of $3^3 = 27$.

4. **Uranium-235 fission**[1]**.** A typical example of an induced fission process in U-235 is:

$$\mathrm{n} + {}^{235}\mathrm{U} \rightarrow {}^{236}\mathrm{U} \rightarrow {}^{137}\mathrm{I} + {}^{96}\mathrm{Y} + 3\mathrm{n} \tag{3}$$

where the ^{236}U exists only momentarily before breaking up into two heavy fragments. Like the famous Oppenheimer, calculate the actual fission energy released (in units of MeV) from this process using the following atomic masses:

$m(^{235}\mathrm{U}) = 235.0439231$ u

$m(^{137}\mathrm{I}) = 136.91787084$ u

$m(^{96}\mathrm{Y}) = 95.91589779$ u

Solutions: The excess energy is given by the energy equivalence of the difference between masses on the left-hand side of the equation and those on the right-hand side:

$$\begin{aligned} E &= \{[m_\mathrm{n} + m(^{235}\mathrm{U})] - [m(^{137}\mathrm{I}) + m(^{96}\mathrm{Y}) + 3m_\mathrm{n}]\}c^2 \\ &= [(1.008664904\ \mathrm{u} + 235.0439231\ \mathrm{u}) - (136.91787084\ \mathrm{u} + 95.9191589779\ \mathrm{u} \\ &\quad + 3 \times 1.008664904\ \mathrm{u})] \times 931.494\ \mathrm{MeV/u} \\ &= 179.6\ \mathrm{MeV} \end{aligned} \tag{4}$$

[1] From Riddle Me These worksheet

5. **More Uranium fission!**[2] What is the mass of a sample of ^{235}U that, when undergoing fission, would produce the same amount of thermal energy as 1 metric ton of coal?

   Note: The thermal energy content of one ton of coal is 3.1×10^{10} J and use your answer from the last question for the energy released in each ^{235}U fission.

   Solutions: The energy released in each fission is 180 MeV$\times(1.6\times10^{-13}$ J/MeV$) = 2.88\times10^{-11}$ J. Thus the number of fissions necessary to produce the same energy as one tonne of coal is

$$\frac{3.1 \times 10^{10}\ \text{J}}{2.88 \times 10^{-11}\ \text{J}} = 1.08 \times 10^{21}\ \text{fissions} \tag{5}$$

   We will need this many ^{235}U atoms to fission, and they will have a mass of

$$\frac{1.08 \times 10^{21}}{6.02 \times 10^{23}\ \text{mol}^{-1}} \times 235\ \text{g/mol} = 0.42\ \text{g} \tag{6}$$

6. **Nuclear-powered world?** Let's estimate how many years of energy the world has left if we switched 100% of our energy consumption to current-gen nuclear power.

   We start with the following assumptions[3]:

   - The current global energy consumption is about 600 Quads (or 2×10^{20} J) per year.
   - 1 kg of U-235 yields around 8×10^{13} J of thermal energy.
   - Natural uranium is only about 0.7% U-235.
   - It's currently estimated that the global uranium reserve is 6×10^{9} kg (6 million tonnes).

   Solution: Let's first figure out how much U-235 fuel we need per year. To get 2×10^{20} J/year, we divide the total energy by the energy density:

$$\frac{2 \times 10^{20}\ \text{J}}{8 \times 10^{13}\ \text{J/kg}} = 2.5 \times 10^{6}\ \text{kg} \tag{7}$$

   Next, we calculate how much natural uranium we must mine. Since only 0.7% of uranium is U-235, we need to mine

$$\frac{2.5 \times 10^{6}\ \text{kg}}{0.007} = 3.5 \times 10^{8}\ \text{kg} \tag{8}$$

[2] From Riddle Me These worksheet

[3] I got these numbers from ChatGPT and did not verify them, so please do not take them seriously. They should, however, be good for an order-of-magnitude estimation.

of natural uranium per year. At this rate, we will burn through the uranium reserve in

$$\frac{6 \times 10^9 \text{ kg}}{3.5 \times 10^8 \text{ kg/year}} \approx 17 \text{ years} \tag{9}$$

This seems short, but note that we are only considering U-235. If we take advantage of the breeder reactors and the Pu-239 they produce, the time it takes to run out of nuclear energy resources would be much longer.